

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing** *Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time Chatbot for Customer Support (AI Application) (2024)

Developed and maintained a scalable AI-powered chatbot leveraging large language model (LLM) inference APIs to provide instant, accurate customer query responses using company knowledge base. Architected a vector database with ChromaDB and integrated OpenAI embeddings to implement a retrieval-augmented generation (RAG) pipeline, ensuring high reliability and performance of the inference services. Collaborated across teams to optimize LLM inference workflows on cloud infrastructure, resulting in a 35% reduction in human agent workload and significant improvements in response time.

Tech: LangChain, OpenAI API, ChromaDB, Python, LLM Integration

Customer Sentiment Analysis for E-commerce (NLP) (2024)

Built a robust NLP pipeline to analyze customer sentiment from social media and e-commerce platform reviews, involving data preprocessing, feature extraction, and classification to drive actionable insights for product enhancement. Employed scalable machine learning models optimized for performance within cloud-native environments. Leveraged vectorization techniques and logistic regression to deliver an 89% accurate sentiment classification, facilitating targeted improvements in product quality, aligned with best practices for scalable AI solutions.

Tech: Python, NLTK, TF-IDF, Logistic Regression, Cloud ML Pipelines

Predictive Maintenance Model for Manufacturing (2024)

Developed a predictive maintenance solution using machine learning to forecast industrial equipment failures by processing time-series IoT sensor data. Leveraged feature extraction pipelines and trained Random Forest models deployed and orchestrated using cloud infrastructure with a focus on scalability and reliability. Applied rigorous evaluation metrics and Kubernetes-based orchestration for seamless integration and operation of inference services, leading to a 28% reduction in unexpected downtime and substantial operational cost savings.

Tech: Python, Random Forest, Sensor Data Processing, Time-Series Modeling, Kubernetes

Customer Churn Prediction for Telecom Industry (2024)

Designed and deployed a machine learning pipeline for predicting customer churn utilizing extensive telecom datasets, enabling proactive retention strategies. Employed advanced data preprocessing, feature engineering, and class imbalance handling techniques to build scalable models including XGBoost, optimized via GridSearchCV. Integrated inference models with cloud-based infrastructure using containerized services and Kubernetes orchestration to ensure performance and scalability, achieving an ROC-AUC of 0.87 and reducing churn by 18%.

Tech: Python, Pandas, NumPy, Scikit-learn, XGBoost, SMOTE, Kubernetes, Cloud Infrastructure

Courses :

Skills :

